

NORTHERN UTILITIES PROCUREMENT AUTHORITY

(a fictional procuring authority created for assessment purposes)

Wadi Sirhan Independent Sewage Treatment Plant

VOLUME II — TECHNICAL REQUIREMENTS (EXTRACT)

Tender Reference: NUPA/ISTP/2026/014

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Sections 1 to 9 — issued for tender

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SECTION 1 — SCOPE OF THE WORKS

- 1.1 The Project Company shall design, finance, construct, commission, own, operate and maintain the Wadi Sirhan Independent Sewage Treatment Plant and shall transfer the Facility to the Authority at the end of the concession period in the condition required by Volume V.
- 1.2 The Works comprise: the inlet works and screening; primary, secondary and tertiary treatment; disinfection; sludge thickening, digestion and dewatering; odour control; the treated effluent transmission main; the treated effluent storage reservoir; and all associated electrical, control and civil works.
- 1.3 The Project Company shall be responsible for all interfaces with the existing collection network at the two designated tie-in points, including any temporary works required to maintain flows during construction.
- 1.4 The Authority will provide the site free of encumbrance, together with a grid connection point at the site boundary. All other utilities shall be procured by the Project Company.

SECTION 2 — DESIGN BASIS

- 2.1 The Facility shall be designed for a nominal treatment capacity of **120,000 m³/day** average daily flow, measured at the inlet works, at the design year influent characteristics stated in Table 2-2.
- 2.2 The design life of all civil structures shall be not less than fifty (50) years. The design life of mechanical and electrical equipment shall be not less than twenty (20) years or the manufacturer's rated life, whichever is the shorter, with lifecycle replacement provided for in the Project Company's lifecycle plan.
- 2.3 The Facility shall be designed to accommodate a future expansion of 50% of nominal capacity without demolition of permanent structures. Space provision only is required; no expansion works form part of the Works.

Table 2-2 — Design influent characteristics

Parameter	Unit	Average	Peak (design)
BOD ₅	mg/l	320	480
COD	mg/l	640	950
Total Suspended Solids	mg/l	350	520
Total Nitrogen	mg/l	55	80
Total Phosphorus	mg/l	9	14
Temperature	°C	24	34 (max) / 14 (min)

- 2.4 The treated effluent shall at all times comply with the parameters set out in Table 2-4. Table 2-4 is reproduced overleaf from the Environmental Permit issued for the site and is incorporated into this Volume by reference.
- 2.5 The Project Company shall install continuous online monitoring for those parameters identified in Table 2-4 as assessed on a continuous basis, with signals available to the Authority in real time.

TABLE 2-4 — EFFLUENT QUALITY (REPRODUCED FROM THE ENVIRONMENTAL PERMIT)

The following page is reproduced as issued by the permitting authority. In the event of any discrepancy between this reproduction and the Environmental Permit, the Environmental Permit shall prevail.

Table 2-4 Minimum Effluent Quality Parameters at the Point of Discharge

(all values are maxima; compliance assessed as a 30-day rolling average unless stated)

Parameter	Unit	Limit	Basis of assessment
Biochemical Oxygen Demand (BOD5)	mg/l	10	30-day rolling average
Chemical Oxygen Demand (COD)	mg/l	50	30-day rolling average
Total Suspended Solids (TSS)	mg/l	10	30-day rolling average
Total Nitrogen (TN)	mg/l	5	30-day rolling average
Total Phosphorus (TP)	mg/l	1	30-day rolling average
Turbidity	NTU	2	Maximum instantaneous
Faecal Coliforms	MPN/100 ml	2.2	Maximum, any single sample
Residual Chlorine	mg/l	0.5 - 1.0	Continuous at outlet
pH	-	6.0 - 9.0	Continuous at outlet
Oil and Grease	mg/l	1	Maximum, any single sample

Note 1: Where a parameter is not listed above, the limit stated in the Environmental Permit shall apply.

End of reproduction.

SECTION 3 — PROCESS REQUIREMENTS

- 3.1 The treatment process shall be capable of achieving the effluent quality in Table 2-4 under all design influent conditions in Table 2-2. The Authority does not mandate a particular process train.
- 3.2 Where a membrane process is proposed, the Project Company shall demonstrate membrane replacement provisions in the lifecycle plan and shall warrant a membrane life of not less than seven (7) years.
- 3.3 Disinfection shall be by chlorination with provision for dechlorination, or by ultraviolet irradiation with a validated dose, or by a combination. Ozonation alone is not acceptable.
- 3.4 Odour control shall achieve not more than 5 OU/m³ at the site boundary, assessed as a 98th percentile hourly value.
- 3.5 Noise at the nearest sensitive receptor shall not exceed 55 dB(A) by day and 45 dB(A) by night.

SECTION 4 — HYDRAULIC DESIGN AND RELIABILITY

- 4.1 The hydraulic profile shall accommodate the flows in Table 2-6 without surcharge of any structure and without reliance on pumped bypass.

Table 2-6 — Hydraulic design criteria

Ref	Criterion	Value	Unit
2-6.1	Average daily flow	120,000	m ³ /day
2-6.2	Peak hourly flow (design)	7,500	m ³ /h

Ref	Criterion	Value	Unit
2-6.3	Minimum night flow	1,800	m ³ /h
2-6.4	Storm flow to be passed to full treatment	3 x dry weather flow	—
2-6.5	Inlet works design head	1.8	m
2-6.6	Maximum velocity in channels	1.2	m/s

- 4.2 The Project Company shall satisfy itself as to the consistency of the flow figures stated in this Volume and shall raise any inconsistency as a clarification under Volume I Clause 5.1.
- 4.3 Each treatment stage shall be arranged in not fewer than four (4) parallel streams, such that the Facility can meet Table 2-4 with any one stream out of service.
- 4.4 Standby provision for all rotating equipment shall be N+1. Standby power shall be sized to carry the full inlet works, the disinfection stage and the effluent pumping station for not less than seventy-two (72) hours.
- 4.5 The transmission main shall be provided with surge protection designed for the full range of operating scenarios including simultaneous pump trip.

SECTION 5 — TRANSMISSION MAIN AND CIVIL WORKS

- 5.1 The treated effluent transmission main shall be approximately eighteen (18) kilometres in length between the Facility outlet chamber and the delivery point shown on Drawing 03-C-114.
- 5.2 The main shall be sized for the peak hourly flow in Table 2-6 with a maximum velocity of 2.0 m/s and a minimum residual head of 15 m at the delivery point.
- 5.3 The main shall be of ductile iron to the relevant international standard, internally lined and externally protected, with a minimum nominal diameter of DN1200. Alternative materials will be considered only where supported by a whole-life cost comparison.
- 5.4 Three (3) road crossings and one (1) wadi crossing shall be executed by trenchless methods. Crossing permits shall be obtained by the Project Company.
- 5.5 All permanent civil structures shall be designed for a seismic event with a return period of 475 years and for a flood with a return period of 100 years.

SECTION 6 — SCADA, CONTROL AND CYBER SECURITY

- 6.1 The Facility shall be provided with a SCADA system capable of unattended operation of the whole Facility, with a permanently manned central control room.
- 6.2 The industrial control system shall be segregated from any business network by a demilitarised zone with unidirectional data flow to the Authority's monitoring interface.
- 6.3 The Project Company shall implement and maintain an information security management system covering the operational technology environment, and shall submit its security architecture for the Authority's review prior to commissioning.
- 6.4 All process data shall be retained for not less than seven (7) years and shall be made available to the Authority on demand in a non-proprietary format.

SECTION 7 — TESTING, COMMISSIONING AND PCOD

- 7.1 The Project Company shall carry out factory acceptance testing, site acceptance testing, and integrated process commissioning in accordance with a commissioning plan approved by the Authority.
- 7.2 PCOD shall not be certified until the Facility has completed a continuous reliability run of thirty (30) days at not less than 90% of nominal capacity, during which every parameter in Table 2-4 has been met.
- 7.3 A failure of any Table 2-4 parameter during the reliability run shall restart the run in full.

- 7.4 The Project Company shall provide as-built documentation, operation and maintenance manuals, and an asset register in the Authority's standard format not later than sixty (60) days after PCOD.

SECTION 8 — OPERATION, MAINTENANCE AND SPARES

- 8.1 The Project Company shall hold a spare parts inventory sufficient for two (2) years of normal operation for all items with a lead time exceeding twelve (12) weeks.
- 8.2 A computerised maintenance management system shall be implemented before PCOD and shall remain in use for the concession period.
- 8.3 The Project Company shall employ a suitably qualified plant manager and shall maintain a minimum of two (2) certified operators on shift at all times.
- 8.4 Sludge shall be dewatered to not less than 22% dry solids and shall be disposed of or reused in accordance with the applicable regulations. Disposal to landfill of sludge exceeding 500 tonnes per annum shall require the Authority's prior written consent.
- 8.5 A handback condition survey shall be carried out in the final two (2) years of the concession, and any remedial works identified shall be completed before transfer.

SECTION 9 — ENVIRONMENTAL AND SOCIAL

- 9.1 The Project Company shall comply with the Environmental and Social Impact Assessment prepared for the site and available in the data room, and shall prepare an Environmental and Social Management Plan aligned with the International Finance Corporation Performance Standards.
- 9.2 Environmental clearance shall be obtained from the competent national environmental regulator before commencement of construction.
- 9.3 A grievance mechanism accessible to the affected community shall be established before construction and maintained for the concession period.
- 9.4 Construction water shall not be abstracted from the local aquifer. The Project Company shall source construction water from tankered supply or treated effluent.
- 9.5 The Project Company shall report environmental and social performance to the Authority quarterly during construction and annually thereafter.